ZZ-S-666D
April 18, 1991
SUPERSEDING
ZZ-S-666C
April 13, 1976

FEDERAL SPECIFICATION
SQUEEGEES, FLOOR AND WINDOW

This Federal Specification has been approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers floor and window squeegees without handles.

1.2 Classification. Squeegees shall be of the following types, classes, and sizes.

Type I - Floor squeegees, wood or plastic block.
  Class 1 - Heavy duty.
    Size 12 - 12 inch.
  Class 2 - Light duty.
    Size 18 - 18 inch.
    Size 24 - 24 inch.

Type II - (Deleted)

Type III - Window squeegees, metal frame.
  Class 5 - Single blade, one piece frame.
    Size 8 - 8 inches.
    Size 10 - 10 inches.
    Size 12 - 12 inches.
    Size 16 - 16 inches.
    Size 24 - 24 inches.
  Class 6 - Double blade, two piece brass frame.
    Size 10 - 10 inches.
    Size 12 - 12 inches.
    Size 16 - 16 inches.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: General Services Administration, Federal Supply Service, General Products Commodity Center, General Engineering Group (7FXEG), 819 Taylor St., Fort Worth, TX 76102

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FSC 7920

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2. APPLICABLE DOCUMENTS

2.1 The documents referenced in this specification shall be the issues in effect on the date of the invitation for bids or request for proposal unless otherwise specified. These documents form a part of this specification to the extent specified. In the event that there is a conflict between this specification and a document referenced herein, this specification shall take precedence.

Federal Specification:

NN-H-104 - Handles, Wood, ACME Threaded and Tapered End.

Federal Standards:

FED-STD-376 - Preferred Metric Units for Use by the Federal Government
FED-STD-601 - Rubber: Sampling and Testing

Copies of Federal specifications and standards are available from the General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

Copies of Military Specifications and Standards are available from Military Specifications and Standards, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

ASTM standards:

D 573 - Standard Test Method for Rubber Deterioration in an Air Oven.

E 29 - Standard Practice For Using Digits In Test Data To Determine Conformance With Specifications.

Application for copies of ASTM standards should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.

3. REQUIREMENTS

3.1 Component and materials. Materials shall conform to the referenced specifications, the requirements in the referenced figures, and as specified in this document.

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3.1.1 Screws, bolts, and nuts. Screws, bolts, and nuts shall be made of corrosion resistant materials or shall be made of steel which is plated to resist corrosion.

3.1.2 Wood or plastic blocks. Blocks shall be closed grained and straight grained hardwood, laminated wood, compressed molded wood particles, solid plastic, or foam plastic. The finished wood block shall be free from imperfections such as warp, decay, knots, checks, splits, or slivers. The finished plastic block shall be smooth or textured, solid, flash free, and free from any defects in molding. The wood or plastic blocks shall be straight, warp free, and shall conform to the dimensional requirements as specified by figures 1A and 1B. The bottom edge of the block shall be grooved and all corners shall be rounded. The screw holes shall be equally spaced along the length through the approximate center of the groove depth. A hole to permit the tapered end of a handle conforming to type II, size 2 or size 3 of NN-H-104 shall be made in each block. The wood blocks shall be water resistant treated by coating with clear wood sealer after machining and before assembly with blade. The blocks shall be dry to the touch before assembly with blade.

3.1.3 Frames and channels. The metal channels and metal strips of the type III squeegees shall be made of brass and shall conform to the dimensional requirements of figures 2A or 2B as applicable. The overall length of the channels and frames shall be 3/8 inch, plus or minus 1/8 inch, less than the size squeegee required by the contract or purchase order (see 6.2). At the option of the contractor, the long leg of the type III, class 5 channel may be curved as shown by cross-section A of figure 2; the long leg shall measure not less than 5/8 inch in overall length with the shorter leg not less than 5/16 inch in overall length.

3.1.4 Rubber blade material. The rubber blades shall be made of natural or synthetic rubber containing no coloring agent which will stain, bleed, or run. The blades shall not become hard nor soft and shall not show signs of cracking or breaking when tested as specified in 4.5.1. The rubber blades shall conform to the dimensions specified by figures 1 and 2. The blades shall be of the length specified by the contract or purchase order (see 6.2). The blades of the type III, class 5 squeegees shall have a molded rubber bead along one edge to fit inside the frame channel; the bead shall be of the diamet er specified by figure 2A. The blades for all type I squeegees and type III, class 6 squeegees shall be flat, of the thickness and width specified by the applicable figure. The working (exposed) edge shall be straight, free from dirt inclusions, cutter marks, or serrations. When replacement blades are specified, the blade only shall be supplied and it shall be of the length specified by the contract or purchase order (see 6.2). The lengths specified for the blades shall be minimums.

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3.1.5 Handle sockets. Handle sockets for the type III squeegees shall be made of brass and shall conform to the dimensions specified on figure 2. All handle sockets shall have an integral collar not less than 1/4 inch in width. The socket handle shall be tapered to accept the tapered end of a handle conforming to type II, size 2 or size 3 of NN-H-104.

3.2 Construction.

3.2.1 Type I. The floor squeegee shall be as specified by figure 1A or 1B as applicable, and shall be a wood or plastic block (see 3.1.2) with a groove along one edge and a rubber blade (see 3.1.4) secured in the groove by screws (see 3.1.1). The blocks shall be equipped with a tapered hole for insertion of a wood handle (see 3.1.2). A complete unit, except where the blade only is specified, shall consist of the assembled block and blade.

3.2.1.1 Class 1. The heavy duty squeegees shall conform to the dimensions in figure 1A. The heavy duty floor squeegees shall have the blade held securely in the grooved portion of the block by not less than four equally spaced flathead wood screws. The screws shall be 1-1/4 inch in length, plus or minus 1/16 inch, countersunk even or deeper than the block surface, the screw points shall not project from the opposite face.

3.2.1.2 Class 2. The light duty squeegees shall conform to the dimensions in figure 1B. The light duty floor squeegees shall have the blade held securely in the grooved portion of the block by the number of equally spaced flathead wood screws specified in the figure. The screws shall be 3/4 inch in length, plus or minus 1/16 inch, countersunk even or deeper than the block surface. The screws shall not project from the opposite face.

3.2.2 Type III. The metal frame squeegees shall be as specified by figures 2A or 2B as applicable. The metal frame squeegee shall be equipped with a socket tapered for insertion of a wooden handle (see 3.1.5). A complete unit, except where the blade only is specified, shall consist of the metal frame, socket, and blade.

3.2.2.1 Class 5. Shall conform to figure 2A. The single blade, one-piece frame window squeegee shall be composed of a one-piece metal channel frame (see 3.1.3) with the round beaded edge of the rubber blade (see 3.1.4) inserted into the channel, the construction shall allow the removal and replacement of the blade without th e use of other than simple tools. Deformation of the channel shall not be used as a means of retention of the blade. The handle socket, as specified in 3.1.5, shall be attached to the channel by not less than two bolts or screws. As an alternate construction, the socket may be secured by one bolt or screw when movement of the socket is not permitted by some design consideration. The threaded end of any bolts used shall not extend more than 1/8 inch from the nut.

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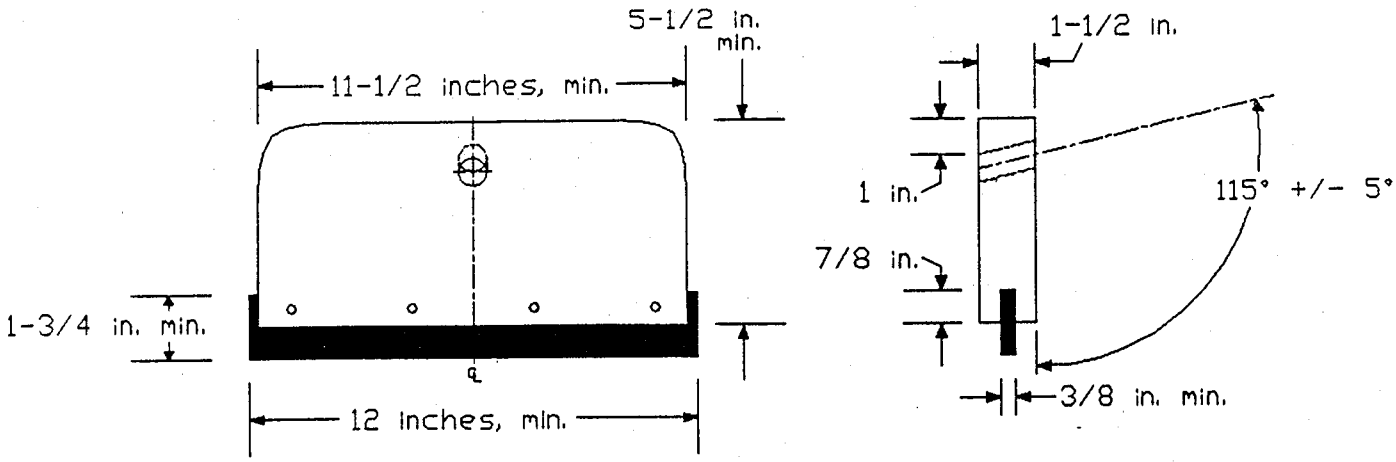
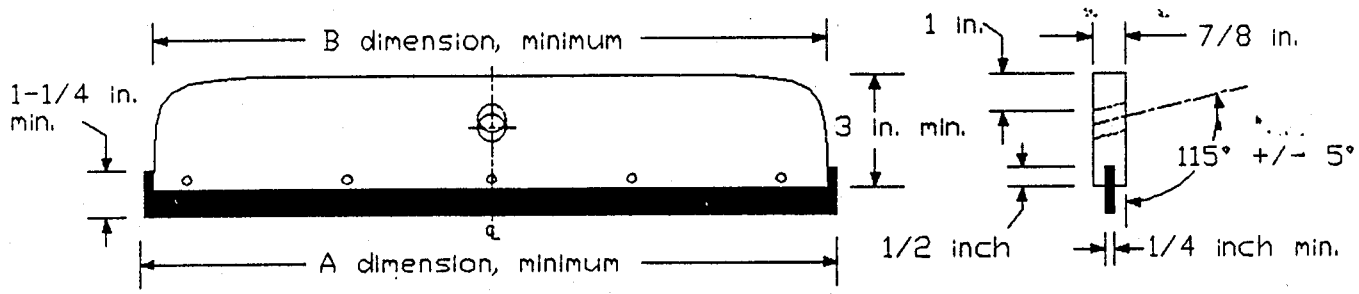


Figure 1 A, Type I, Class 1



Size	A	B	# Screws
18	18	17-1/2	5
24	24	23-1/2	7

Figure 1 B, Type I, Class 2

Note: All dimensions w/o specified tolerances shall have the tolerance of plus or minus 1/8 inch.

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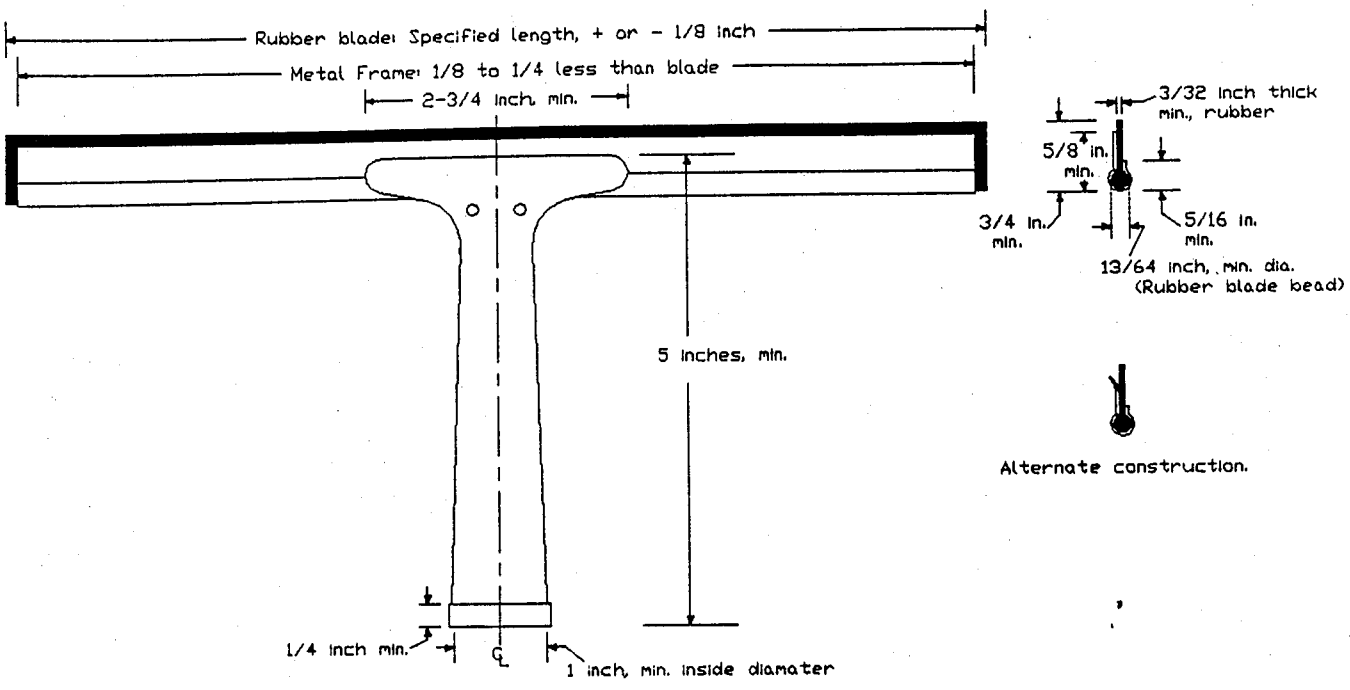


Figure 2 A, Type III, Class 5

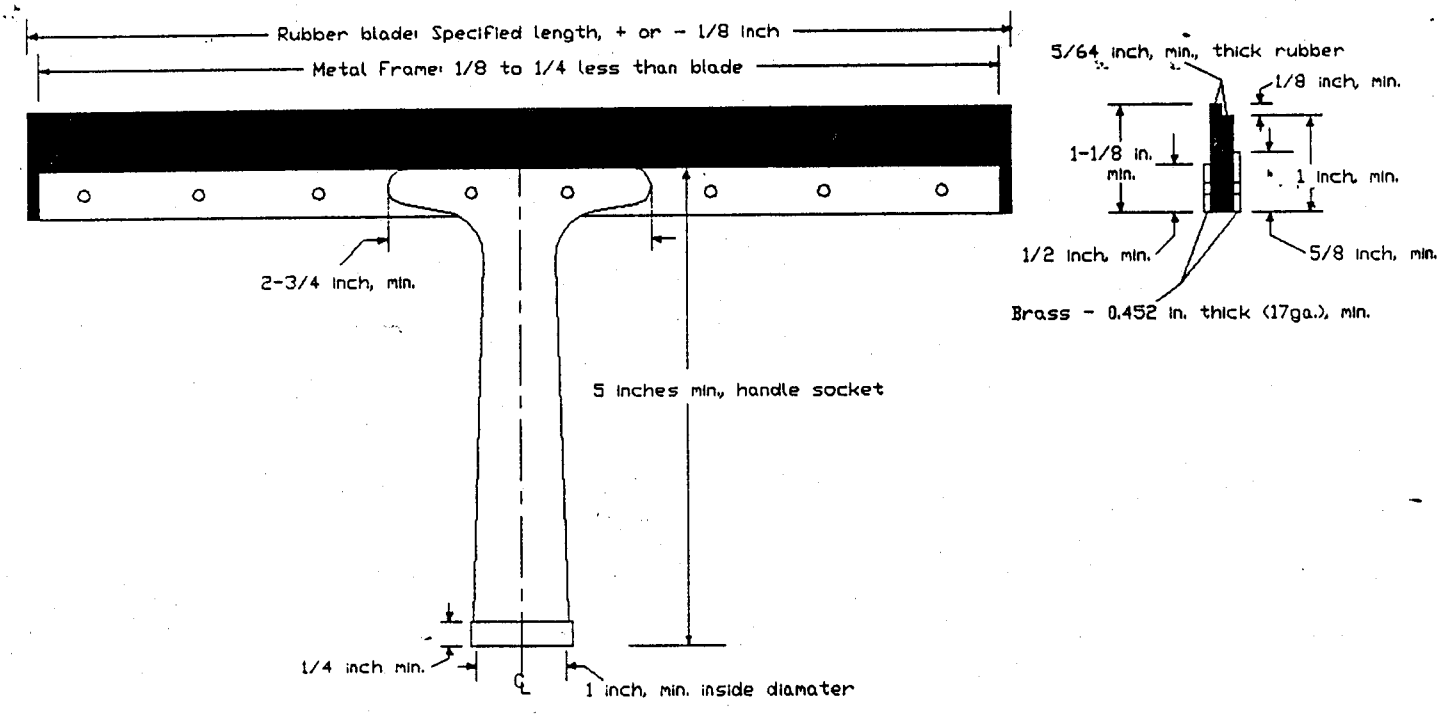


Figure 2 B, Type III, Class 6

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3.2.2.2 Class 6. Shall conform to figure 2B. The double blade, two-piece frame window squeegees shall have a 2-ply rubber blade (see 3.1.4) secured in between the two pieces of the metal strip frame (see 3.1.3) by not less than 6 equally spaced screws or bolts of not less than 1/8 inch diameter. The bolts or screws shall extend through the first strip, both rubber blades, and the other metal strip, and shall not protrude more than 1/8 inch beyond the nuts used or the metal strip if no nuts are used. The handle socket, as specified in 3.1.5, shall be attached to the strips by not less than two of the above bolts or screws, or by not less than two separate bolts or screws. When flathead screws are used they shall be countersunk into the metal strips so that they are flush with the surface of the metal strip.

3.4 Performance requirements.

3.4.1 Type I squeegees. There shall be no visible evidence of surface water when tested as specified in 4.5.2.

3.4.2 Type III squeegees. There shall be no visible evidence of surface water or streaks when tested as specified in 4.5.3.

3.5 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.6 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this specification are met.

3.7 Workmanship. The finished squeegees shall be stra ight, clean, well made, and free from any defects which may affect appearance or serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified in this specification. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections or tests set forth in the specifications where such action is deemed necessary to assure that supplies and services conform to the prescribed requirements. The contractor is responsible for insuring that components and materials used were manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Certifications of conformance. For the tests specified in 4.5.1, component certificates of conformance are acceptable in lieu of lot by lot testing of the end item. These certificates of conformance shall be based on component testing of the lot of incoming material, and may be performed by the component supplier. Each lot of the incoming component item shall either be certified by the component supplier or be tested by the contractor as basis for the end item certification. All certificates submitted shall have test reports supporting them attached.

4.3 Sampling. Sampling shall be in accordance with MIL-STD-105, the inspection level and the Acceptable Quality Level (AQL) shall be as specified.

4.4 Visual and dimensional examination. Visual and dimensional examination of the end item shall be done on a lot by lot basis. The lot shall be all squeegees of the same type, class, and size submitted for inspection at one time, the sample unit shall be one complete squeegee. Where the blade only is specified, the sample unit shall be one complete blade. The inspection level for visual examination shall be I with AQL 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units. The inspection level for dimensional examination shall be S-3 with an AQL of 4.0. Tolerance limits specified are absolute limits as defined in ASTM E 29

4.4.1 Visual examination. Each sample unit shall be examined for the visual defects as listed in table I.

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Table I, Classification of defects, visual examination.

Examine	Defect	Classification Major	Minor
Finish			
Metal	Evidence of orange peel, wrinkles, drops, or streaks.	X	
	Dirt, grit, or foreign material imbedded, evidence of corrosion.	X	
Wood	No water resistant treatment.	X	
	Any warps, slivers, decay, knots, checks, roughness or unsanded parts.		X
Workmanship			
All	Sharp burr, splinter or sliver, rough or sharp edges.		X
	Blade has chips, splits or deterioration, dirt inclusions, cutter marks, or serrations.	X	
	Blade is wavey or not straight.	X	
Type I	Block corners not rounded, holes not true, groove not straight, groove not centered, holes not distributed evenly along length of block.	X	
	No handle hole or hole not tapered.	X	
	Blade not held securely in block or the number screws less than specified.	X	
	Screws not countersunk, or heads or points extend beyond face of block.		X
	Screws do not pass through center of groove.		X
	Screws not evenly spaced.	X	
Type III	Less than two screws securing handle socket to frame, except where alternate construction is furnished.	X	
	Alternate construction allows handle to rotate.	X	
	Handle socket is loose.	X	
	Flathead screws not countersunk when furnished.		X
	Screws extend more than 1/8 inch beyond nut, or strip when no nut used.	X	

4.4.2 Dimensional examination. The samples shall be inspected for compliance with the dimensional requirements.

4.5 Testing. Certification as a result of component testing, specified by 4.2, is acceptable for the tests specified in paragraph 4.5.1. The test specified by paragraph 4.5.2 shall be performed on a lot by lot basis.

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4.5.1 Rubber blade test. For component testing the sample unit shall be 1 foot of material, the inspection shall be S-2, failure of any test shall fail the lot. For end item testing the sample unit shall be one blade, the inspection shall be S-2, the AQL shall be 2.5 defects per 100 units. Each sample shall be put through the series of tests in the order that the tests are given as follows:

4.5.1.1 Soaking test. The samples shall be immersed in clear, clean water at 170 degrees Fahrenheit, plus or minus 5 degrees Fahrenheit, for 15 minutes. The blades shall be removed and examined for compliance with 3.1.4.

4.5.1.2 Accelerated aging test. After the soaking test the blades shall be tested in accordance with ASTM D 573 at 160 degrees Fahrenheit (plus or minus 2 degrees F) for 94 hours. The blades shall be examined for compliance with 3.1.4.

4.5.1.3 Bending test. After the accelerated aging test each of the samples shall be secured with a 1/2 inch rods held tightly to each side in a direction perpendicular to the long axis of the blade, and bend tested as follows: Bend to a 90 degree angle over the rod on one side, return to the straight direction, and bend to a 90 degree angle over the rod on the other side; this shall be repeated for a total of 25 cycles without removing or loosening the sample from the securing device. Each of the samples shall be examined for compliance to 3.1.4.

4.5.2 Wiping test for type I floor squeegees. Four squeegees from each lot of the same type, class and size submitted for inspection shall be selected for this test. The squeegee to be tested shall have a suitable handle securely attached. A smooth, flat, floor surface of 10 square feet or greater shall be washed with clean, clear water. With the blade of the squeegee wet, and while the floor is still wet, the floor shall be wiped with not more than 2 pushing or forward strokes of the squeegee in the same area. The floor shall then be examined for compliance with 3.4.1.

4.5.3 Wiping test for type III window squeegees. Four squeegees from each lot of the same type, class and size submitted for inspection shall be selected for this test. An uncoated glass window pane of 3 square feet or greater, in a vertical position, shall be washed with clean, clear water. With the blade of the squeegee wet, and while the pane is still wet, the pane shall be wiped with not more than 2 downward strokes of the squeegee in the same area. The pane shall then be examined for compliance with 3.4.2.

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4.6 Examination for preparation for delivery. A random sample of unit, intermediate and shipping containers (as applicable) shall be selected from each lot and examined for conformance with the preservation, packaging, packing, labeling and marking required in the contract or purchase order. The Inspection level shall be S-2, AQL 6.5 expressed in terms of defects per hundred units.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The packaging, packing, and marking shall be as specified in the contract or purchase order.

6. NOTES

6.1 Intended use. The squeegees covered by this specification are used for wiping floors and windows dry.

6.2 Ordering data. Purchasers should select the preferred options permitted here and include this information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, Style, and Size, of squeegee (see 1.2).
(c) If replacement blade only is required.
(d) Packaging, packing, and markings required.

6.3 Part numbering. Purchase descriptions using this document shall use the following part numbering system:

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- Type: I = 1, III = 3
- Class: 1 = A, 2 = B, 5 = C, 6 = D
- Size: 8 inch = 8, 10 inch = 10, 12 inch = 12, 15 inch = 15, 16 inch = 16, 18 inch = 18, 24 inch = 24

When present, an "R" preceding the type designato r indicates a replacement blade only.

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6.4 National stock numbers. The following NSN's have been identified as using this document:

NSN	Part Number (see 6.4)
7920-00-224-8339	ZZS666-1A12
7920-00-965-4873	ZZS666-1B18
7920-00-530-5740	ZZS666-1B24
7920-00-267-4639	ZZS666-3C8A
7920-00-577-4747	ZZS666-3C10
7920-00-577-4748	ZZS666-3C12
7920-00-267-4638	ZZS666-3C15
7920-00-267-2972	ZZS666-3C16
7920-00-577-4744	ZZS666-3D10
7920-00-577-4745	ZZS666-3D12
7920-00-577-4746	ZZS666-3D16
7920-00-577-4754	ZZS666-R3C10
7920-00-577-4755	ZZS666-R3C12
7920-00-577-4756	ZZS666-R3C16

MILITARY INTERESTS

NONE: DoD has determined that no military activity has an official interest in this Federal Specification.

Preparing Activity
GSA-FSS

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